

Christopher Jacob “C.J.” Ticknor

Ph.D. Candidate | Department of Chemical and Environmental Engineering | University of Cincinnati

4515 Smith Rd, Apt 13
Cincinnati, OH 45212

Phone 937-654-8425

E-mail ticknocj@mail.uc.edu

LinkedIn ticknorcj

Website <https://ticknorcj.github.io/>

ORCID 0009-0000-1653-9834

Github ticknorcj

Chemical engineering PhD researcher focused on the optimization of molecular diagnostics for point-of-care applications. Interested in applying innovative design, fabrication techniques, and simulation to deliver robust and optimized diagnostic platforms to address critical bottlenecks in global health.

Education

University of Cincinnati

Aug. 2022 - Present

Doctor of Philosophy | Chemical Engineering

- Dissertation: “Eddy Current-Driven Thermal Control in Centrifugal Microfluidics for Integrated Diagnostics”
- Advisor: Dr. Aashish Priye

The Ohio State University

Aug. 2015 – Dec. 2020

Bachelor of Science | Chemical Engineering

Publications

Peer-Reviewed Journal Articles:

- [1] Miriam Knutson, Sasanka P. Weerakoon, **C. J. Ticknor**, Benjamin M. Yavitt, Aashish Priye; *Development and Implementation of a Low-Cost 3D-Printed Rotational Viscometer for Rheology and Fluid Mechanics Education*. J. Chem. Educ. 11 March 2025; 102 (3): 1138–1145

Presentations

Talks:

- [1] **C Ticknor**, A Priye. (2025, November). *Eddy Current-Driven Thermal Control in Centrifugal Microfluidics for Integrated Diagnostics*. Paper presented at the AIChE Annual Meeting, Boston, MA.
- [2] **C Ticknor**, A Priye. (2024, October). *Modeling and Optimization of Eddy Current-Induced Microfluidic Heating for Lab-on-a-Disc Platforms*. Paper presented at the AIChE Annual Meeting, San Diego, CA.

Posters:

- [1] **C Ticknor**, A Priye. (2025, April). *Optimization of Eddy Current-Induced Microfluidic Heating for Lab-on-a-Disc Platforms*. Poster presented Research + Innovation Week Tech Expo, Cincinnati, OH.

Research Experience

Graduate Research Assistant

Aug. 2022 - Present

Priye Lab | University of Cincinnati

Principal Investigator: Dr. Aashish Priye

- Iteratively designed and constructed a custom experimental system using microelectronics, CAD, 3D printing, CNC, and laser cutting to demonstrate the non-contact heating via induced electrical eddy currents
- Optimized the conditions of the test system using experimental data to deliver applications such as wax valving and actuation of biochemical reactions
- Developed analytical and finite element method COMSOL simulations to effectively capture the coupled physics of induced electromagnetic eddy currents and time-dependent heat transfer
- Mentored undergraduate researchers by teaching additive and subtractive manufacturing and supervised progress to yield deliverables such as reports, papers, and posters

Graduate Research Assistant

Jan. 2024 - Present

P&G Digital Accelerator | University of Cincinnati / Procter & Gamble

Principal Investigators: David Eike, Joanna Jaworska

- Tested and verified calculation of physicochemical properties related to skin permeation using COSMOtherm
- Developed scripts for automated parameter calculation, SQL database storage, and input file generation for the skin permeation model DigiSkin
- Extended the capabilities of the skin permeation model DigiSkin, a gPROMS FormulatedProducts model, to account for 3D geometry and capture age-related effects
- Simulated the application of various acids on skin to drive key performance insights such as pH control, proton flux, and depth-dependent concentration profiles

Teaching Experience

University of Cincinnati

- Teaching Assistant, CHE Laboratory (CHE 5037) Aug. 2022 - May 2023
Demonstrated pilot-scale unit operations to undergraduate students using key concepts from curriculum to enable student-led experiments on subsequent lab sessions

Academic and Volunteer Service

Tutor, Woodward Career Technical High School

Apr. 2026 - Present

Honors and Awards

Eagle Scout, Boy Scouts of America

Aug. 2014

Skills

Programming & Data Tools: Python, MATLAB, C++, Mathematica, JMP

Simulation & Process Modeling: COMSOL, ChemCAD, COSMOtherm, gPROMS FormulatedProducts

CAD / Design / Engineering: Fusion360, SolidWorks, FreeCAD, ChemDraw

Electronics & Automation: Arduino/microcontrollers, KiCAD, LabVIEW

Fabrication & Prototyping: CNC machining, 3D printing, laser cutting, PCB design, soldering / electronics assembly

Productivity & Office Tools: Microsoft Office, LaTeX

Design & Media Tools: Davinci Resolve

Computer Systems: Microsoft Windows, Linux/Unix

Industry Experience

Research and Development Technician

Mar. 2021 - Jul. 2022

Nagase ChemteX America | Delaware, Ohio

- Developed and tested chemical products for applications including solar panels, adhesives, and small electronics, supporting product evaluation from concept through final formulation.
- Collaborated with chemists to design and refine product formulations, contributing to end-to-end development and iterative problem-solving.
- Assisted in experimental design by selecting test methods, defining variables, and helping establish workflows for product validation.
- Defined key performance indicators for new products and identified benchmark standards to evaluate performance and guide development decisions.
- Organized and maintained batch history data to support traceability, documentation, and product quality review.
- Tracked project plans, recorded experimental findings, and presented results to internal staff and customers.
- Managed sample storage in compliance with safety requirements to protect personnel, preserve sample integrity, and support lab organization.
- Cleaned, maintained, and recalibrated laboratory equipment to ensure accurate, reliable testing and continued operational readiness.

Quality Intern

May 2020 - Aug. 2020

Hohman Plating | Dayton, Ohio

- Interpreted procedures and components for improvement in the Quality Management System to support the business's quality objectives
- Interpreted customer quality system requirements for industrial processing and inspection specifications
- Managed oven charts for metal plating processes, issued periodic testing work orders, managed test panel data and inventory
- Utilized various data management platforms, including WorkFlo, TrueChem, Microsoft Dynamics GP, Microsoft Access, and Excel

Language

English: Native

Last updated in August 2026